

## Multiple Choice

1. **Answer:** A

*Options:* A) True, True; B) False, False; C) True, False; D) False, True

*Note:* Correct option: A - True, True

2. **Answer:** A

*Options:* A) Supervised learning; B) Unsupervised learning; C) Clustering; D) None of the above

*Note:* Correct option: A - Supervised learning

3. **Answer:** B

*Options:* A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.

*Note:* Correct option: B - Improve the optimisation algorithm being used for error minimisation.

4. **Answer:** A

*Options:* A) Transform data to zero mean; B) Transform data to zero median; C) Not possible; D) None of these

*Note:* Correct option: A - Transform data to zero mean

5. **Answer:** B

*Options:* A) pure; B) not pure; C) useful; D) useless

*Note:* Correct option: B - not pure

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'polynomial in D'.

7. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

8. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'O(D)'.

## Long Form Response

11. **Answer:** `def remove_odd(str 1): | return str 2`

*Note:* Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def armstrong_number(number): | return True | return False`

*Note:* Students should provide a detailed explanation referencing algorithm steps.

*End of Answer Key*